

Elementary Matrices

* Lemma 1.1 Matrix multiplication is associative i.e. $A.(B.C) = (A.B).C$

* Lemma 1.2 If $P = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $Q = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix}$ are matrices s.t. $\exists R = P.Q$ then $R = \begin{bmatrix} AA' + BC' & AB' + BD' \\ CA' + DC' & CB' + DD' \end{bmatrix}$
if # columns of A and C = # rows of A' and B'

* Defn 1.3 A matrix A is called invertible if $\exists B$ s.t. $A.B = B.A = I$

* Lemma 1.4 A is invertible $\iff A^T$ is invertible

* Lemma 1.5 If A, B and C are matrices s.t. A and B are invertible and $C = A.B \implies C$ is invertible

* Lemma 1.6 Elementary Matrices $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}$
 $\begin{cases} \rightarrow \text{Effect of left and right multiplication} \\ \rightarrow \text{They are invertible} \end{cases}$

* Defn 1.7 Reduced row echelon form (rref):

- ① The first non zero element in a row, i.e. pivot, is 1. All the elements above and below the pivot are zero
- ② If row i and row j are two rows of rref s.t. $j > i$, then if both rows have pivot then pivot of row j is to the left of row i
- ③ If row i is all zero then all the rows below it are all zero.

* Lemma 1.8

A (square matrix)

invertible

$\searrow$

$\text{rref}(A) = I$

non invertible

$\searrow$

$\text{rref}(A) = \begin{bmatrix} P & Q \\ 0 & 0 \end{bmatrix}$ and also $\exists$ invertible matrix M s.t. $\text{rref}(A) = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} M$

* Theorem 1.9

If A, B, C are matrices s.t. $C = A.B$
 C is invertible $\iff$ Both A and B are invertible

Hint Use 1.8 and 1.5

* Lemma 1.10

A is an invertible matrix $\iff A$ is product of elementary matrices

* Defn 1.11

Determinant : $\mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ s.t

① $\det(I) = 1$

② $\det\left(\begin{bmatrix} a \\ \vdots \\ ap+ba \\ \vdots \\ a \end{bmatrix}\right) = a \det\left(\begin{bmatrix} 1 \\ \vdots \\ p \\ \vdots \\ 1 \end{bmatrix}\right) + b \det\left(\begin{bmatrix} 1 \\ \vdots \\ 1 \\ \vdots \\ 1 \end{bmatrix}\right)$, i.e. det is linear in rows

③ If two rows of A are same then $\det(A) = 0$

* Lemma 1.12

If E is an elementary matrix and A are square matrices then

$$\det(EA) = \det(E) \det(A)$$

* Theorem 1.13

If A and B are square matrices s.t $\exists A, B$ then

$$\det(AB) = \det(A) \det(B)$$

Hint 1.12, 1.10

* Lemma 1.14

If A is a square matrix then $\det(A) = \det(A^T)$

* Theorem 1.15

If A, P, Q are square matrices s.t $A = \begin{bmatrix} P & 0 \\ 0 & Q \end{bmatrix}$ then $\det(A) = \det(P) \det(Q)$

Hint (i) P or Q is non invertible

(ii) P and Q are invertible

$$f(B) = \frac{\det \begin{bmatrix} B & 0 \\ 0 & Q \end{bmatrix}}{\det(Q)}$$

$f(B)$ satisfies 1.11 and $\exists$ a det rule.

* Note 1.16

Discussions upto here can be extended to complex matrices

* Defn 1.17

S_n : Symmetric group: Group of permutations of n elements

Cyclic notation for elements of S_n

* Defn 1.18

GL_n : General linear group

$$GL_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t. } \exists B \text{ s.t. } AB = BA = I \}$$

* Defn 1.19

SL_n : Special linear group

$$SL_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t. } AB = BA = I, \det(A) = 1 \}$$

* Defn 1.20

O_n : Orthogonal group

$$O_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t. } A \cdot A^T = A^T \cdot A = I \}$$

* Defn 1.21

SO_n : Special Orthogonal group

$$SO_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t. } A \cdot A^T = I \text{ and } \det(A) = 1 \}$$

* Lemma 1.22

Let A, B, C be square matrices and x a column vector s.t

$$Ax = B \quad \text{--- (i)}$$

$$CAx = CB \quad \text{--- (ii)}$$

If C is invertible $\Rightarrow$ solution space of (i) = solution space of (ii)

If C is non invertible $\Rightarrow$ solution space of (i) is a subspace of the solution space of (ii)

Groups

* Defn 2.1 Law of composition on set S is $f: S \times S \rightarrow S$

* Def 2.2 $e \in S$ is called identity w.r.t a law of composition if $e \cdot a = a \cdot e = a \quad \forall a \in S$
 Caution: A set might not have an identity element w.r.t a law of composition

* Defn 2.3 Transposition is an element of S_n that switches only two elements

* Defn 2.4 Group: A set G together with a law of composition $(\cdot)$ s.t

① $(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \forall a, b, c \in G$

② $\exists e \in G$ s.t $e \cdot a = a \cdot e = a \quad \forall a \in G$

③ if $a \in G \exists a^{-1} \in G$ s.t $a \cdot a^{-1} = a^{-1} \cdot a = e$

* Example 2.5 S_3 : Symmetry group of 3 elements

$x = (123) \quad y = (12)$

$x^3 = 1 \quad y^2 = 1 \quad yx = x^2y$

$S_3 = \{1, x, x^2, y, xy, x^2y\}$

generators

defining relations

* Defn 2.6 Subgroup H of group G , is a non empty subset s.t

① if $a, b \in H$ and $c = a \cdot b \Rightarrow c \in H$

② if $a \in H \Rightarrow a^{-1} \in H$

* Example 2.7

$$\begin{matrix} & C_n & \\ SO_n & \subset & C_n \\ & \subset & S_n \end{matrix}$$

* Defn 2.8 $H \leq G$ is called normal subgroup, i.e $H \trianglelefteq G$

if $\forall g \in G$ and $h \in H \quad ghg^{-1} \in H$

* Defn 2.9 Order of group G is the number of unique elements in G .

* Defn 2.10 Let U be the subset of group G . Then the group generated by U is defined as the smallest subgroup H of G that contains all elements of U

* Defn 2.11 Cyclic subgroup: The group generated by a single element x of a group
 $\langle x \rangle := \{ \dots, x^2, x^1, e, x, x^2, \dots \}$
 Order of $x :=$ Order of $\langle x \rangle$

* Defn 2.12 Group Homomorphism: $\varphi: G \longrightarrow G'$ s.t.
 $\varphi(a.b) = \varphi(a).\varphi(b) \quad \forall a, b \in G$

Image of $\varphi := \{ \varphi(g) : g \in G \}$
 kernel of $\varphi := \{ g \in G : \varphi(g) = e' \}$

* Lemma 2.13 (i) $\varphi(e) = e'$
 (ii) $\varphi(a^{-1}) = \varphi(a)^{-1}$
 (iii) if $H \subset G \implies \varphi(H) \subset G'$

* Lemma 2.14 (i) kernel is a normal subgroup of G
 (ii) Image is a subgroup of G'

* Lemma 2.15 A homomorphism φ is injective i.e one to one if kernel of φ is $\{e\}$

* Defn 2.16 Isomorphism: A bijective group homomorphism

* Defn 2.17 Automorphism: An isomorphism from group to itself

* Defn 2.18 $a, b \in G$ are called conjugates if $\exists g \in G$ s.t $b = g a g^{-1}$

* Defn 2.19

Let $H \leq G$ and $a \in G$ then

$$\text{Left coset of } a := aH = \{a \cdot h \in G : h \in H\}$$

↳ All the left cosets have the same size

* Theorem 2.20

Let $H \leq G$ then

$$[G] = \underbrace{[H]}_{\text{order of } H} \underbrace{[G:H]}_{\text{\# of left cosets}}$$

Hint: $a \sim b$ if $\exists h \in H$ s.t. $a = b \cdot h$ is an equivalence relation $\Downarrow$

$$[G] = [H][G:H] \quad \leftarrow \begin{array}{l} \text{Left cosets are non intersecting} \\ 2.19 \end{array}$$

* Corollary 2.21

Let $\varphi: G \rightarrow G'$ be a group homomorphism,

$$[G] = [\text{kernel}] [\text{Image}]$$

* Correspondence: Theorem 2.22 Let $\varphi: G \rightarrow G'$ be a surjective group homomorphism

$$(i) \quad \left[\begin{array}{l} \text{Subgroups of } G \text{ that} \\ \text{contain the kernel} \end{array} \right] \xleftrightarrow[\text{map}]{\text{bijective}} \left[\begin{array}{l} \text{Subgroups of} \\ G' \end{array} \right]$$

$$(ii) \quad \{ H \triangleleft G \text{ and } K \subset H \} \xleftrightarrow[\text{map}]{\text{bijective}} \{ H' \triangleleft G' \}$$

* Notes: surjection of φ is needed for

(a) the correspondence to be onto

(b) $\varphi(H) \triangleleft G'$ where $H \triangleleft G$ and $K \subset H$

* Defn 2.23

Quotient Group: Let $H \triangleleft G$, $\bar{a} := aH$ left coset of $a \in G$ then

$G/H := \{ \bar{a} : a \in G \}$ with the law of composition $\bar{a} \cdot \bar{b} = \overline{a \cdot b}$ has the group structure

Further $\varphi: G \rightarrow G/H$ where $\varphi(a) = \bar{a}$ is a group homomorphism

Hint: Prove that law of composition is valid for arbitrary representative element and then it carries over the group structure

* First
Isomorphism
Theorem 2.24

Let $\varphi: G \rightarrow G'$ is a surjective homomorphism and K be its kernel, then
 $\exists$ an isomorphism f b/w G/K and G'

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & G' \\ \pi \searrow & & \nearrow f \\ & G/K & \end{array}$$

$$f(\bar{a}) := \varphi(a)$$

* Lemma
2.25

If w_1 and w_2 are subgroups of G , then the following are also subgroups

$$w_1 \cap w_2 := \{g \in G : g \in w_1 \text{ and } g \in w_2\}$$

$$w_1 + w_2 := \{g \in G : g = a \cdot b \text{ or } g = b \cdot a \text{ where } a, b \in w_1 \cup w_2\}$$

* Usually $w_1 + w_2 \neq w_1 \cup w_2$

* Lemma
2.26

All the subgroups of $\mathbb{Z}^+$ (additive group of integers) have the form
 $n\mathbb{Z}$ where $n \in \mathbb{Z}$.

Hint: Use well ordering property of $\mathbb{Z}$

* GCD
Defn 2.27

If $a, b \in \mathbb{Z}$, then $d \in \mathbb{Z}$ is called gcd if

(i) $d > 0$

(ii) d divides a and b

(iii) if $c \in \mathbb{Z}$ divides a and b then c divides d

* Lemma 2.28

If $a, b, d \in \mathbb{Z}$ st d is gcd of a and $b \implies d\mathbb{Z} = a\mathbb{Z} + b\mathbb{Z}$

* Defn 2.29

$a, b \in \mathbb{Z}$ are called relatively prime if $\text{g.c.d}(a, b) = 1$

* Defn 2.30

Let $p \in \mathbb{Z}$ st if p divides $a \cdot b \implies p$ divides either a or b

* Lemma 2.31

$p \in \mathbb{Z}$ is prime $\iff \exists n \in \mathbb{Z}$ st $1 < n < p$ and n divides p .

* L.C.M If $a, b \in \mathbb{Z}$, then $l \in \mathbb{Z}$ is called gcd if

defn 2.32

(i) $l > 0$

(ii) a and b divides l

(iii) if a and b divides $c \in \mathbb{Z}$ then l divides c

* Lemma 2.33 If $a, b, l \in \mathbb{Z}$ s.t l is l.c.m of a and $b \implies l\mathbb{Z} = a\mathbb{Z} \cap b\mathbb{Z}$

* Lemma 2.34 $\text{g.c.d}(a, b) \times \text{l.c.m}(a, b) = a \cdot b$

Prove (i) $a \cdot b / d$ is divisible by l using 2.32

(ii) $l \cdot d$ is divisible by $a \cdot b$ using $d = ma + nb$

* Defn
2.48

Modular arithmetic

Congruence Relation $a \equiv b$ if $a - b$ is divisible by n

$f: \mathbb{Z}^+ \rightarrow \mathbb{Z}/n\mathbb{Z}$ is group homomorphism with $f(a+b) = f(a) + f(b)$

further (i) $f(a \cdot b) = f(a) \cdot f(b)$

(ii) $f(a) (f(b) + f(c)) = f(a) f(b) + f(a) f(c)$

Vector Space

- * Defn 3.1 A field F is a set together with two laws of composition, i.e. addition ⁽⁺⁾ and multiplication ^(*) st
- (i) $(F, +)$ is an abelian group with identity denoted by 0
 - (ii) $(F, \cdot)$ is an abelian group, where $F^* = \{F - \{0\}\}$. Identity is denoted by 1.
 - (iii) $\forall a, b, c \in F \quad a \cdot (b + c) = ab + ac$

- * Lemma 3.2 $F_p = \mathbb{Z}/p\mathbb{Z}$ where $p \in \mathbb{Z}$ is prime is a field.
- Hint: Proof by contradiction. Let $p = a \cdot b$ with $1 < a, b < p$
show that closure fails for F_p^*

- * Lemma 3.3 F_p^* is an group of order $p-1$ and is cyclic, i.e.
- $$F_p^* = \{1, \alpha, \alpha^2, \dots, \alpha^{p-2}\} = \{\bar{1}, \bar{\alpha}, \dots, \overline{p-1}\}$$
- The generator α of F_p^* is called primitive root modulo p .

- * Defn 3.4 The characteristic of a field is the order of $\langle 1 \rangle \subset (F, +)$

- * Lemma 3.5
- characteristic of a field

Finite

↙

It is a prime number

Infinite

↘

then it is called zero

Hint: Same as 3.2

- * Defn 3.6 A vector space V over a field F is a set together two laws, i.e. vector addition and scalar multiplication s.t
- (i) V is an abelian group with scalar addition. Identity element is 0_V
 - (ii) if $a \in F, v \in V$ then $a \cdot v \in V$
 - (iii) if $v \in V \quad 1 \cdot v = v$
 - (iv) if $u, v \in V$ and $a, b \in F \implies$

$$a(u+v) = au + av$$

$$(a+b)v = av + bv$$

$$a \cdot (b \cdot v) = (a \cdot b) \cdot v$$

- * Lemma 3.7 let $v \in V, 0_F \in F \implies 0_F \cdot v = 0_V$

* Defn 3.8

A subspace of a vector space V over field F is a non empty subset s.t

- ① if $u, v \in V$ then $u+v \in V$
- ② if $a \in F, v \in V$ then $a \cdot v \in V$

* Lemma 3.9

If W_1 and W_2 are subspaces of V then

- ① $W_1 \cap W_2$ is a subspace of V
- ② $W_1 + W_2$ is a subspace of V , where $W_1 + W_2$ is the group generated by $W_1 \cup W_2$ (2.25)

* Defn 3.10

Ordered Set $\begin{cases} \rightarrow \text{Order of elements matter} \\ \rightarrow \text{Repetition of elements is not allowed.} \end{cases}$

* Defn 3.11

Span of an ordered set $S = \{v_1, v_2, \dots\}$ with $v_i \in V$ is the subspace generated by linear combinations of v_i 's, i.e.

$$\text{Span}(S) := \left\{ \sum_i a_i v_i \text{ where } a_i \in F, v_i \in V \right\}$$

* Defn 3.12

A vector space V is called finite dimensional if $\exists$ a finite set S s.t
 $V = \text{Span}(S)$

* Defn 3.13

An ordered set $S = \{v_1, v_2, \dots, v_n\}$ with $v_i \in V$ is called linearly independent if
 $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0_V \iff a_i = 0 \text{ where } 1 \leq i \leq n$

* Defn 3.14

An empty set $\emptyset$ is linearly independent
 $\text{Span}(\emptyset) = \{0_V\}$

* Defn 3.15

A basis B of vector space V is an subset s.t

- ① B is linearly independent
- ② $\text{Span}(B) = V$

* Defn 2.16 The dimension of a finite dimensional vector space is defined to be number of vectors in its basis

* Lemma 3.17 Let L be a linearly independent set of finite dimensional V , S be a set of V s.t. $\text{Span}(S) = V$ then $\dim(\text{Span}(L)) \leq \dim(\text{Span}(S))$

hint: Proof by contradiction $\dim(\text{Span}(L)) > \dim(\text{Span}(S))$

* Lemma 3.18 Let V be a finite dimensional vector space and $[B]$ and $[B]'$ are its two basis associated by $[B]' = [B]P$ $\begin{matrix} \xrightarrow{\quad} \\ \downarrow \end{matrix}$ $\begin{matrix} P \text{ is invertible} \\ \text{base change matrix} \end{matrix}$

If $v \in V$ then $\alpha = [B]'x' = [B]x$ $\xrightarrow{\quad}$ $\begin{matrix} \text{coordinate vector of } v \text{ w.r.t } [B] \end{matrix}$

$x = Px'$

* Defn 3.19 The two subspaces U_1 and U_2 of V are called linearly independent if $U_1 \cap U_2 = \{0\}$

* Defn 3.20 A vector space V is called direct sum of subspace $U_1, U_2, \dots, U_k$ if

- ① $U_1, U_2, \dots, U_k$ are linearly independent
- ② $V = U_1 + U_2 + \dots + U_k$

It is denoted by $V = U_1 \oplus U_2 \oplus \dots \oplus U_k$

* Lemma 3.21 If W_1 and W_2 are subspace of V then $\dim(W_1) + \dim(W_2) = \dim(W_1 + W_2) + \dim(W_1 \cap W_2)$

Linear Transformation

* Defn 4.1 A linear transformation T is a map from vector space V to vector space W , both over the same field F s.t

$$\textcircled{1} T(u_1 + u_2) = T(u_1) + T(u_2) \quad \forall u_1, u_2 \in V$$

$$\textcircled{2} T(av) = aT(v) \quad \forall v \in V, \forall a \in F$$

* Defn 4.2
Kernel of $T := \{u \in V : T(u) = 0_W\}$
Image of $T := \{T(u) : u \in V\}$

* Lemma 4.3
Kernel is a subspace of V
Image is a subspace of W

* Lemma 4.4
 $\dim(\text{Kernel}) + \dim(\text{Image}) = \dim(V)$
nullity rank

Hint $[B] := \{ \overbrace{a_1, a_2, \dots, a_k}^{\text{basis of kernel}}, b_1, \dots, b_m \}$
Show $\{T(b_1), T(b_2), \dots, T(b_m)\}$ is a basis of Image

* Lemma 4.5
Let $T: V \rightarrow W$ be a linear transformation, P a subspace of V
then $T(P)$ is a subspace of Image

* Defn 4.6
Let $T: V \rightarrow W$ be a linear transformation, $[B], [C]$ be the basis of V and W respectively
Then the matrix of linear transformation T w.r.t $[B]$ and $[C]$ is a matrix A s.t
 $T([B]) = [C]A$

Lemma
If $x \in F^n, y \in F^m$ be the coordinate vectors of $u \in V$ and $T(u) \in W$ w.r.t $[B]$ and $[C]$ respectively
then $y = Ax$

Lemma
If $[B']$ and $[C']$ are another sets of basis of V and W s.t
 $[B'] = [B]P \quad [C'] = [C]Q$

then A' , the matrix of linear transformation T w.r.t $[B']$ and $[C']$ is given by
 $A' = Q^{-1}AP$

* Defn 4.7

An Isomorphism b/w two vector spaces V and W over the same field F is a

bijective linear transformation $T: V \rightarrow W$

Example: A n dim vector space V over field F is isomorphic to the vector space F^n .

* Lemma

4.8

If A is an square matrix then

$$Ax = 0 \iff A \text{ is invertible}$$

iff only if $x = 0$

Proof (i) A is invertible $\Rightarrow$ let $Ax = 0 \Rightarrow A^{-1}Ax = A^{-1}0 \Rightarrow x = 0$

(ii) let $Ax = 0$ only for $x = 0 \Rightarrow$ columns of A are linearly independent

$[A]$ is a basis of $\mathbb{R}^n$, $[I]$ is another basis of $\mathbb{R}^n \therefore \exists P$ s.t. $[A]P = [I]$
invertible

* Lemma

4.9

let $A_{m \times n}$ be a matrix and $E_{n \times n}$ be an invertible matrix, then

$$\dim(\text{ColSpace}(A)) = \dim(\text{ColSpace}(EA))$$

Hint use 4.4, 4.5

* Lemma

4.10

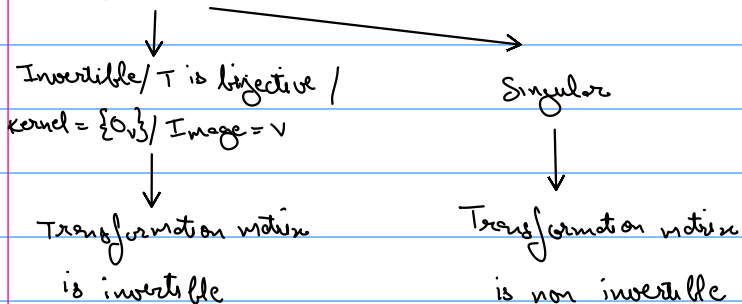
let $A_{m \times n}$ be a matrix then

$$\dim(\text{ColSpace}(A)) = \dim(\text{RowSpace}(A))$$

Use 1.8, 4.9

* Defn 4.11

A linear operator T is a linear transformation from vector space V to itself.



* Lemma

4.12

Let $T: V \rightarrow V$ be a linear operator, then the following conditions are equivalent

(i) $V = \text{Kernel}(T) \oplus \text{Image}(T)$

(ii) $\text{Kernel}(T) \cap \text{Image}(T) = \{0_v\}$

(iii) $\text{Kernel}(T) + \text{Image}(T) = V$

These don't mean T is invertible

E.g: $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Hint: Use 4.4, 3.21

* Lemma 4.13 Let $T: V \rightarrow V$ be a linear operator, $[B], [B']$ be two different basis of V associated by $[B'] = [B]P$ and A, A' be the transformation matrix w.r.t $[B]$ and $[B']$ then

$$A' = P^{-1}AP, \text{ further}$$

A and A' are called similar matrices as they represent the same linear transformation

* Defn 4.14 Let $T: V \rightarrow V$ be a linear operator, then a subspace W of V is called T -invariant if $T(W) \subset W$

* Lemma 4.15 Let T be an linear operator on a vector space V , W be an T -invariant subspace of V , $\{w_1, w_2, \dots, w_k\}$ is the basis of W , $[B] := \{w_1, w_2, \dots, w_k, v_1, v_2, \dots, v_n\}$ be a basis of V , $f: V \rightarrow V/W$ be a map defined by $f(v+W) := \bar{v}$

(i) $\bar{v}_1 + \bar{v}_2 = \overline{v_1 + v_2}$
(ii) $a\bar{v}_1 = \overline{av_1}$ } are independent of the choice of representative element.

(iii) V/W along with (i) and (ii) form a vector space

(iv) $f: V \rightarrow V/W$ is a linear transformation

(v) $[\bar{B}] := [\bar{v}_1, \bar{v}_2, \dots, \bar{v}_n]$ is an basis of V/W .

(vi) $\varphi: V/W \rightarrow V/W$ given by $\varphi(\bar{v}) := f(T(v))$ is an linear operator

(vii) The transformation matrix of T w.r.t $[B]$ has the form $A = \begin{bmatrix} P_{k \times k} & Q_{k \times n-k} \\ 0_{(n-k) \times k} & R_{(n-k) \times (n-k)} \end{bmatrix}$

(viii) The transformation matrix of φ w.r.t $[\bar{B}]$ is $R_{n \times n}$

* Lemma 4.16 An eigen vector v of a linear transformation $T: V \rightarrow V$, where V is a vector space over field F , is an

(i) non zero vector

(ii) $T(v) = \lambda v$ where $\lambda \in F$

* Note 4.17 An eigen vector is an one dimensional T -invariant subspace

* Lemma 4.18 Let V be a vector space over field F , $T: V \rightarrow V$ an linear operator, $\lambda \in F$ then

$\lambda I - F: V \rightarrow V$ is an operator

* Lemma 4.19

Let $0 \in V$ be an eigen vector of $T: V \rightarrow V$ with eigen value λ , then $\downarrow$

characteristic polynomial of T $\left\{ \det(I - \lambda I) = 0 \right\}$ where A is the matrix of T $\leftarrow \lambda I - T$ is singular $\leftarrow 0 \neq 0$ is in the null space of $\lambda I - T$

* Lemma 4.20

- (i) The characteristic polynomial of $T: V \rightarrow V$ is independent of the choice of basis
- (ii) Similar matrices have the same characteristic polynomial
- (iii) The coefficients of characteristic polynomial is independent of the choice of basis
- (iv) Eigen values are roots of the characteristic polynomial

* Lemma 4.21

Eigen vectors corresponding to distinct eigen values are linearly independent

Hint: Proof by induction.

* Lemma 4.22

If $T: V \rightarrow V$ has n distinct eigen values then $\exists$ basis $[B]$ of V s.t the transformation matrix of T is diagonal, i.e. $\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$

* Schur's Lemma 4.23

Let $T: V \rightarrow V$ be a linear operator,

If the characteristic polynomial $\Rightarrow \exists$ basis $[B]$ of V s.t the transformation matrix of T is upper diagonal $\begin{bmatrix} p & q \\ 0 & r \end{bmatrix}$

Hint Use 4.20(iv), 4.17, 4.15 (vi), (vii) to prove upper diagonal

Proof it is not always diagonalizable :- let v_1 be the eigen vector of $T: V \rightarrow V$, V_1 the span of v_1 , $f: V \rightarrow V/V_1$ where $f(v_1) = \bar{v}$

Let v_2 be the eigen vector of $\varphi: V/V_1 \rightarrow V/V_1$ $\varphi(\bar{v}) := f(T(v))$

$$v_2' = v_2 + cv_1 \quad T(v_2) = \lambda_2 v_2 + nv_1$$

$$T(v_2') = T(v_2) + T(cv_1) = \lambda_2(v_2' - cv_1) + c\lambda_1 v_1 = \lambda_2 v_2' + (n + c(\lambda_1 - \lambda_2))v_1$$

then if $\lambda_1 \neq \lambda_2$ we can diagonalize

if $n=0$ we can diagonalize

if $\lambda_1 = \lambda_2$ and $n \neq 0$ we cannot diagonalize

* Defn 4.24 A generalized eigen vector x , with eigen value λ and exponent d of $T: V \rightarrow V$ is a vector s.t

$$① \quad x \neq 0_v$$

$$② \quad (T - \lambda I)^d x = 0$$

$$③ \quad \nexists (T - \lambda I)^k x = 0 \implies k > d$$

* Lemma 4.26 Let $x \in V$ be a generalized eigen vector x of $T: V \rightarrow V$ with eigen value λ and exponent d ,

$$u_i := (T - \lambda I)^i x \quad \text{with} \quad 0 \leq i \leq d-1$$

$$[B] := \{u_1, u_2, \dots, u_{d-1}\}$$

$$X := \text{span}([B])$$

then

(i) B is linearly independent

(ii) X is T invariant

(iii) $T([B])$ has the form $\begin{bmatrix} \lambda & & 0 \\ 1 & \lambda & \\ & \ddots & \ddots \\ 0 & & 1 & \lambda \end{bmatrix}_{d \times d}$

Proof Hint (i) Show that any nontrivial combination $[B]$ is a generalized eigenvector $\therefore \neq 0_v$

* Lemma 4.25 A generalized eigen vector x of $T: V \rightarrow V$ has unique eigen value λ and exponent d

Proof Hint: Use binomial expansion

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

* Jordan Decomposition Theorem 4.26 Let T be a linear operator on a finite dimensional complex vector space V . Then

$\exists$ a basis $[B]$ of V s.t the transformation matrix of T w.r.t $[B]$ has the

Jordan form.

Proof: We will prove

as they together lead to the

basis $[B]$ w.r.t which the

transformation matrix of T

has the jordan form.

$$\begin{cases} ① & V = V_1 \oplus V_2 \oplus \dots \oplus V_n \\ ② & V_i = \text{span}([u_{i,0}, u_{i,1}, \dots, u_{i,d_i-1}]), \quad 1 \leq i \leq n \\ ③ & u_{i,j} = (T - \lambda_i I)^j u_{i,0}, \quad 1 \leq j \leq d_i-1 \end{cases}$$

generalized eigen vector with eigen value λ_i and index d_i

We know that ①, ②, ③ are true for $\dim V$.

We will use principle of induction, i.e. assuming the statement is true for complex vector space of dimensions 1 to $n-1$ prove it for n dim complex vector space.

* Let $T: V \rightarrow V$ be a linear operator on n dim complex vector space V . Since the characteristic polynomial factors into linear terms in complex field $\exists$ an eigen vector v with eigen value λ .

If w.r.t basis $[z]$ T has jordan form then so does $T - \lambda I$. Therefore for simplicity we work with $T - \lambda I$ and refer to it as T , then

$T: V \rightarrow V$ is a linear operator and its nullspace is non-trivial.

Let K_i be the kernel and U_i be the image of $T^i: V \rightarrow V$. Since V is finite dimensional let $m \in \mathbb{Z}$ be the smallest number s.t.

$$K_i = K_j \quad \text{and} \quad U_i = U_j \quad \forall i, j \geq m$$

then let $v \in K_m \cap U_m$, since $T^m v = 0$ and $\exists x \in V$ s.t. $T^m x = v$

$$T^m v = 0 \Rightarrow x \in K^m \therefore v = T^m x = 0$$

$$\text{Using 4.12} \quad V = K_m \oplus U_m$$

$\dim(U_m) < n \therefore \exists$ jordan generators. If $\dim(K_m) < n$ the proof is complete otherwise T is an idempotent operator

* $T: V \rightarrow V$ be an idempotent operator, prove $\exists$ a basis $[z]$ s.t. T has jordan form.

Let K be the kernel of T and U be the image of T . Since $\dim(U) < n \exists$ jordan generators for it i.e.

$$U = U_1 \oplus U_2 \cdots \oplus U_r \quad \text{and}$$

$$U_i = \text{Span}([u_{i,0}, u_{i,1}, \dots, u_{i,d_i-1}]), \quad 1 \leq i \leq r,$$

$$u_{i,j} = (T)^j u_i, \quad 1 \leq j \leq d_i - 1$$

generalized eigen vector with eigen value 0 and index d_i

Let $w_i \in V$ s.t. $T w_i = 0$ then w_i is a generalized eigen vector with eigen value 0 and index $d_i + 1$

$$\text{Let } W_i = \text{Span}([w_{i,0}, w_{i,1}, \dots, w_{i,d_i}]) \quad \text{where} \quad w_{i,j} = T^j w_i$$

$$W = W_1 + W_2 \dots + W_n$$

Let $w \in W_n \cap W_k$ then $Tw \in V_n \cap V_k$ since $V_n \cap V_k = 0$

$$w \in K \text{ Since } W_n \cap K = \text{Span}(u_{n,d_n}) = \text{Span}(v_{n,d_n-1})$$

$$\text{and } W_k \cap K = \text{Span}(u_{k,d_k}) = \text{Span}(v_{k,d_k-1})$$

Since v_{n,d_n-1} and v_{k,d_k-1} are linearly independent

$$w = 0$$

$$\therefore W = W_1 \oplus W_2 \oplus \dots \oplus W_n$$

Let R = subspace of K that is not in W

Augment basis of R to W to get $[B]$ s.t. T has Jordan form w.r.t it.

* lemma 4.27 If T is a linear operator on a finite dimensional complex vector space V .

If for some $n \in \mathbb{Z}$, T is the identity operator i.e. $T^n = I \Rightarrow T$ is diagonalizable